



Novatra Solar Investment DSS

AI-Powered Rooftop Solar
Investment Decision Support

Team Members:

Aaditya Partha Sarathy - 230701001
Bharath Kumar M - 230701053



Abstract

Novatra Solar DSS is a full-stack, AI-driven decision support system that helps Indian households make informed rooftop solar investment decisions. The system combines a two-stage ML tariff forecasting pipeline (trained on CEA electricity data, 2012–2025) with MNRE ALMM certified module data to size the optimal solar system, estimate costs including subsidies, and project 25-year ROI under three scenarios — Pessimistic, Base, and Optimistic. Risk indicators (breakeven CAGR, cost-overflow stress, loan-stress test) and a clear BUY / HOLD / AVOID recommendation are delivered through a clean React frontend backed by a FastAPI engine, empowering users to make confident, data-driven solar investment decisions.

Problem Statement

Electricity tariffs in India rise unpredictably — households lack reliable forecasts.

- ✓ What technology and system size fits my roof and budget?
- ✓ Will the investment pay back in my lifetime?
- ✓ What are the risks if tariffs don't rise as expected?
- ✓ How do subsidies (MNRE / state) affect my net cost?

Current tools only provide static estimates — no ML forecasting, no risk analysis, no personalised recommendation.



Existing Systems

Tata Power Solar Calculator

- ✓ Estimates system size and cost
- ✗ Static calculation only — no forecasting
- ✗ No tariff scenario analysis
- ✗ No technology recommendation from ALMM

Loom Solar Estimator

- ✓ Simple, quick estimate
- ✗ No future tariff trend modelling
- ✗ No ML-based forecasting pipeline
- ✗ No risk indicators or decision output

Disadvantages of Existing Systems



Static & Non-Adaptive

No ML forecasting. Tariffs change yearly — these tools cannot model future scenarios.



Limited Personalisation

Generic estimates — not tailored to roof area, shading, state subsidies, or loan options.



No Scenario Comparison

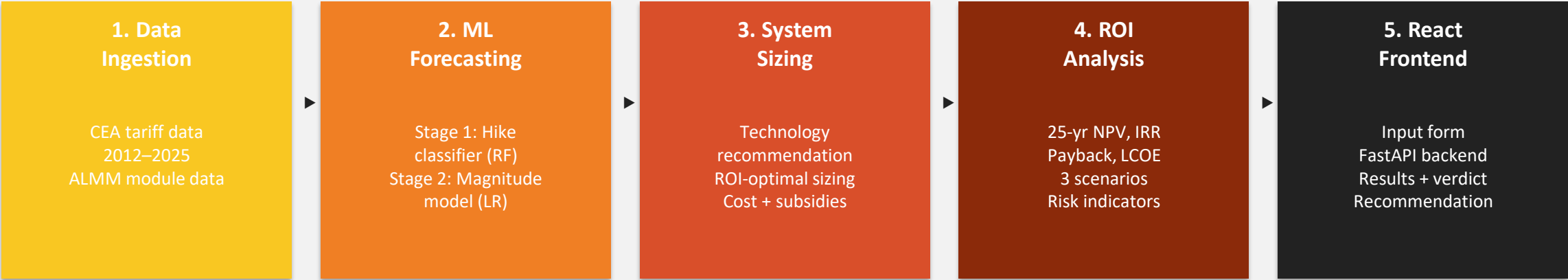
Cannot evaluate Pessimistic / Base / Optimistic scenarios or quantify payback risk.



No Risk or ROI Visibility

No NPV, IRR, LCOE, breakeven CAGR, or loan-stress analysis provided.

Proposed System — Novatra Solar DSS



Key Inputs:

- State, monthly kWh, budget (₹), usable roof area (m²)
- Optional: shading loss fraction (vendor-measured)
- Optional: loan amount, rate, tenure

Key Outputs:

- System size, panel tech, inverter, cost breakdown
- NPV / IRR / LCOE / Payback — 3 scenarios, 25 years
- STRONG BUY / BUY / CONDITIONAL BUY / AVOID verdict

ML Tariff Forecasting Pipeline

Stage 1 — Hike Classifier (RandomForest)

- Predicts: will a state hike its tariff next year? (binary)
- Features: lag_1, yoy_change, years_since_last_hike, state cluster
- Hard rule: frozen ≥ 4 years $\rightarrow$ always predict hike
- Optimised threshold via walk-forward validation (2019–2021)
- Trained on CEA data 2012–2021, evaluated on 2022–2025 actuals

Stage 2 — Magnitude Model (LinearRegression)

- Predicts: by how much will the tariff increase?
- Only runs when Stage 1 probability > 0.45 or hard rule fires
- lag_1 explains $\sim 93\%$ of variance; LR beats complex models here
- Never predicts a tariff lower than current (floor enforced)

15-Year Scenario Forecast

- Year 1 (2026): two-stage ML pipeline prediction
- Years 2–15: CAGR compounded from ML year-1 output
- 3 scenarios — Pessimistic / Base / Optimistic
- Cluster-specific CAGR: A=7.5%, B=5.0%, C=3.0%, D=1.5% (base)
- Output: forecast_multiyear_long.csv — DSS-ready format

25-Year ROI Extension

- Years 16–25: cluster CAGR continuation (clearly labelled as assumption)
- Optimistic multiplier: $1.4\times$ base CAGR; Pessimistic: $0.6\times$
- Provides full panel-lifetime NPV, IRR, and payback analysis

Advantages of Proposed System



AI-Driven Tariff Forecasting

Two-stage ML pipeline (RandomForest + LinearRegression) trained on 14 years of CEA data delivers state-specific 15-year tariff forecasts across 3 scenarios.



Real ALMM Module Data

Technology recommendation powered by MNRE ALMM certified module records — efficiency scores, manufacturer coverage, and value-score ranking.



Full Financial Analysis

NPV, IRR, LCOE, payback period (15yr + 25yr), loan EMI modelling, MNRE + state subsidy calculation, and a clear BUY / AVOID verdict.



Quantified Risk Indicators

Breakeven tariff CAGR, cost-overrun stress test (+12%), and loan-stress test (+2pp rate) — risks are named and measured, not ignored.



End-to-End Product

React + FastAPI full-stack application. Users enter 4 inputs and receive a complete investment report in seconds — no spreadsheets needed.



Thank You

Novatra Solar DSS — AI-Powered Rooftop Solar Investment Tool

Aaditya Partha Sarathy | Bharath Kumar M

